ARCUS

INFRASTRUCTURE INTELLIGENCE

PATH 01 / NEW TERRITORY / PROVINCE-LEVEL PRELIMINARY SCREENING

Territory Briefing: Torino

Evidence-based preliminary infrastructure intelligence for technical meetings, early design discussion and due diligence. ARCUS supports screening; it does not replace site-specific checks or design verification.

REPORT ID	PROJECT CONTEXT	DOMINANT DRIVER	EVENTS
ARCUS-P01-20260603-torino	Road crossing	Hydraulic	42
SPATIAL LEVEL	**ANALYSIS TYPE**	**VERSION**	**GENERATED BY**
Province level	Territory Briefing	ARCUS Path 01 v1.0	ARCUS Professional

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Executive Summary

ARCUS identifies 42 documented collapse precedents in the selected province and translates the Step 3 hazard-overlay state into operational next steps for road crossing interventions. The dominant territorial signal is **Hydraulic exposure**; the leading historical cause is **Hydraulic**.

Recommended use: province-level preliminary screening before design, due diligence or site-specific investigation.

Dominant exposure	Historical signal	Priority use
Hydraulic exposure / score 97	**Hydraulic represents 98% of the selected evidence.**	**Identify road segments intersecting flood-prone corridors or hydraulic concentration areas.**

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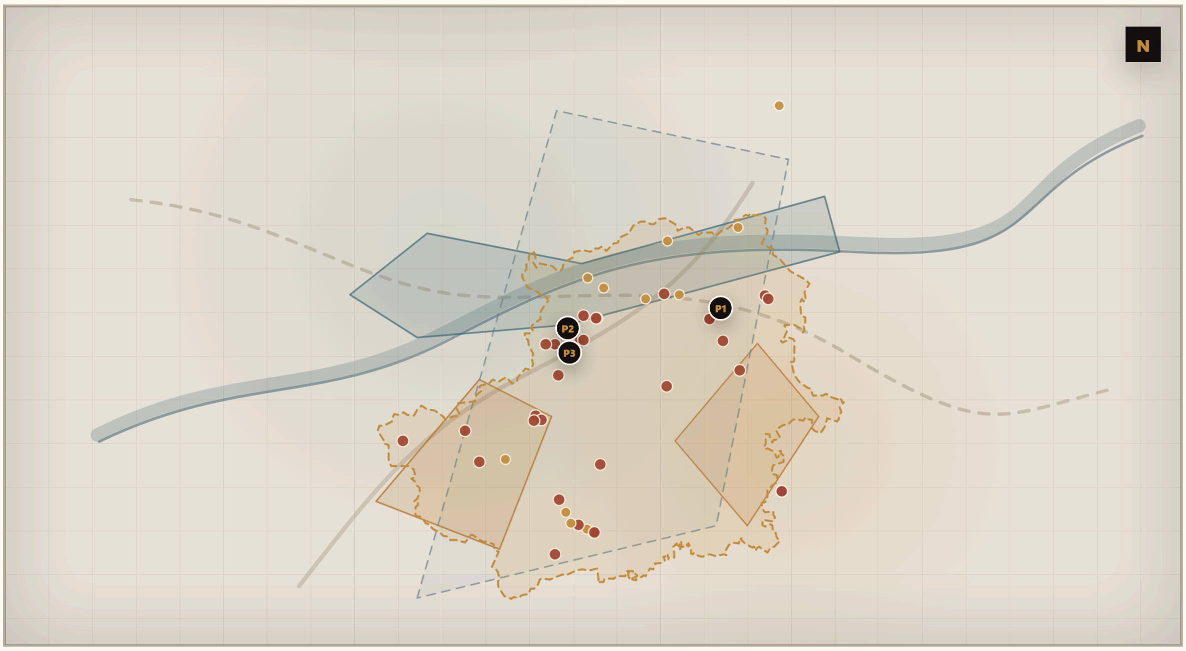
Selected Province & Real Atlas Map



TOTAL COLLAPSE · TRIGGERED EVENT · HYDRAULIC OVERLAY · LANDSLIDE OVERLAY · SEISMIC CONTEXT

> Territorial screening map. Not for design scale. P1/P2/P3 markers identify priority events listed in the table below.

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How to Read This Report

ARCUS combines historical collapse records, public hazard context, source-backed evidence and comparable precedents. The output is intended for preliminary screening and planning of follow-up checks, not for structural safety certification.

SELECTED PROJECT CONTEXT

Road crossing

Road corridor, culvert, minor bridge, embankment, drainage crossing or road-continuity problem: network interruption, drainage bottlenecks, embankments and access continuity.

DOMINANT EXPOSURE

Hydraulic exposure

41 matched ARCUS events. Public overlays are used as screening context and must be followed by site-specific checks.

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Key Indicators

ARCUS EVENTS	TOTAL COLLAPSE	TRIGGERED	SOURCES
42	30	41	125

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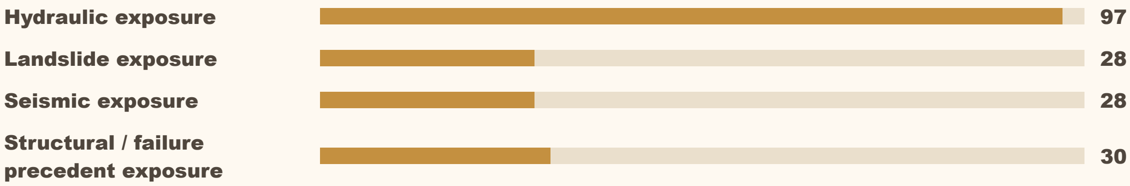
Hazard / Exposure Context

The Step 3 view combines hydraulic WMS, landslide WMS, seismic context and ARCUS historical collapses. The bars below summarise the province-level exposure signal used by the recommendation engine.



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Operational Meaning

PRIMARY FOCUS

Road continuity under flood scenarios; culvert and minor crossing capacity; drainage bottlenecks.

For road crossing interventions in the selected province, ARCUS indicates that preliminary attention should focus on road continuity under flood scenarios and related checks because the dominant indicator is hydraulic exposure. This is a province-level screening signal and must be translated into site-specific checks by qualified professionals.

ATTENTION POINTS

- Road continuity under flood scenarios
- Culvert and minor crossing capacity
- Drainage bottlenecks
- Embankment erosion and local washout
- Debris blockage
- Alternative access routes

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Priority ARCUS Precedents

P1/P2/P3 are shown on the Step 3 map. These records are historical analogues for screening and investigation planning, not direct predictions.

P1	**B00.10.15 - Castellamonte** 2000-10-13 / Hydraulic	TC / High
P2	**B00.10.03 - Ala di Stura** 2000-10-13 / Hydraulic	TC / High
P3	**B00.10.05 - Ala di Stura** 2000-10-13 / Hydraulic	TC / High

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Comparable Precedents

Comparable precedents are selected by severity, trigger, dominant driver, evidence confidence and human impact. For this context, the useful question is not whether a future event is predicted, but which mechanisms and data gaps should be checked before design decisions.

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Interpretation

For road crossing interventions in Torino, ARCUS indicates that preliminary investigation should be framed around hydraulic exposure, documented collapse precedents and the operational consequences implied by the selected project context.

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Immediate Screening Actions

ACTION 1

Identify road segments intersecting flood-prone corridors or hydraulic concentration areas.

ACTION 2

Review culverts, minor bridges and drainage bottlenecks that may concentrate flow or debris.

ACTION 3

Assess potential network interruption and alternative access during extreme rainfall or flood events.

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Data to Request Before Design Decisions

DATA 1
Road drainage layout

DATA 2
Culvert inventory

DATA 3
Road embankment geometry

DATA 4
Flood-prone area mapping

DATA 5
Historical interruption records

DATA 6
Local maintenance records

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Design-Phase Checks

ACTION 1
Translate province-level screening into site-specific checks by qualified professionals.

ACTION 2
Compare planned intervention assumptions with P1/P2/P3 historical precedents.

ACTION 3
Use public WMS overlays as context, not as detailed hydraulic, geotechnical or seismic modelling.

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Recommended Client Path

1. Identify road segments intersecting flood-prone corridors or hydraulic concentration areas.
2. Review culverts, minor bridges and drainage bottlenecks that may concentrate flow or debris.
3. Assess potential network interruption and alternative access during extreme rainfall or flood events.
4. Compare the planned intervention with ARCUS priority precedents around Castellamonte, Ala di Stura, Felleto, Balme, Chialamberto.
5. Request the missing technical data before design or institutional decisions.

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Data Coverage & Limitations

Province-level screening only; not design scale.

ARCUS does not certify structural safety or asset condition.

Public overlays support prioritisation and must be followed by site-specific investigations.

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Methodology Snapshot

ARCUS combines documented collapse events, source reliability, spatial relevance, severity, triggers, cause similarity, territorial hazard context and comparable precedents. The dominant exposure indicator is calculated from Step 3 scores: hydraulic, landslide, seismic and structural/failure precedent exposure.

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Score and Class Legend

P1/P2/P3 identify the top priority historical precedents on the exported Atlas map. Severity TC means total collapse; triggered events indicate an external initiating condition was recorded. Prefer "Failure precedent exposure" unless asset-level structural data are available.

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Short Source Appendix

B24.09.01 / PROTEZIONE CIVILE
Evento meteo-idrologico 05 settembre 2024

B17.06.01 / JOURNAL PAPER
Bridge collapses in Italy across the 21st century: survey and statistical analysis

B08.05.02 / JOURNAL PAPER
Bridge collapses in Italy across the 21st century: survey and statistical analysis

B08.05.02 / REGIONE PIEMONTE
Evento alluvionale 29-30 maggio 2008 – relazione tecnica ufficiale

B05.11.01 / JOURNAL PAPER
Bridge collapses in Italy across the 21st century: survey and statistical analysis

B00.10.03 / ARPA PIEMONTE
Evento alluvionale regionale 13-16 ottobre 2000 – monografia ufficiale

B00.10.03 / JOURNAL PAPER
Bridge collapses in Italy across the 21st century: survey and statistical analysis

B00.10.04 / ARPA PIEMONTE
Evento alluvionale regionale 13-16 ottobre 2000 – monografia ufficiale

> The full source table is provided as a separate supporting CSV export.